1 | Benchmark

1.1 Version 2

I design the following groups of perturbation to test the SQG+1 model. Just to recall, from the SQG simulation obtain

$$\eta_{qg} = \phi^0_{qg}$$

The target field is constructed by

$$\eta_{tar} = \eta_{qg} + \epsilon \mathcal{N}(\phi^0_{qg})$$

Which is essentially QG field adding the cyclogeostrophic adjustment. QG simulation is just a source for a physical streamfunction. The observed field is

$$\eta_{obs} = \eta_{tar} + \eta_{per}$$

Where $\eta_{per} \sim O(\epsilon)$ is the observation error, I design several types of perturbations. All perturbations are normalized so that the RMS of the perturbation matches the Rossby number times the RMS of the target field:

$$\mathrm{rms}(\eta_{per}) = \epsilon \cdot \mathrm{rms}(\eta_{tar})$$

Before normalization, the spatial mean is removed to avoid a DC offset. Given an unnormalized perturbation η_{per}, the final perturbation is

$$\eta_{per} = \frac{\epsilon \cdot \mathrm{rms}(\eta_{tar})}{\mathrm{rms}(\eta_{per} - \langle \eta_{per} \rangle)} \left(\eta_{per} - \langle \eta_{per} \rangle\right)$$

where $\langle \cdot \rangle$ denotes the spatial mean. This ensures every benchmark case starts with the same cost function value, so the comparison isolates the structure of the perturbation rather than its amplitude.

Next Step

Currnetly, the scale of the perturbation is same as the cyclogeostrophic adjustment. That is

$$\mathcal{O}(\eta_{per}) = \mathcal{O}(\mathrm{Ro})$$

What I should actually do is seperate the scale of them two :

- Compute the system Rossby number :

  $$\epsilon = \mathrm{Ro}_{sys} = \frac{\langle u \rangle}{f L_d}$$

  Where $\langle u \rangle$ computes the mean veloctiy as the characteristic velocity scale and L_d is the deformation radius.

- Compute the Rossby number scale for the perturbation :

  $$\mathrm{Ro} = \frac{\xi_{rms}}{f}$$

In the simulation setup, fix ϵ based on the field. Alter the Ro to test influence of the scale of the perturbation. Ro can even be larger than 1.

1.1.1 Group A: Structure-correlated perturbations

These perturbations are deterministic and spatially correlated with the SQG field.

- **A1 — Zero perturbation (control).** Just nothing, the initial loss function for this case should be machine precision.
- **A2 — Frontogenesis-correlated.** Proportional to $|\nabla\psi|^2$, which is large at fronts and strain regions:
$$\eta_{per} = \left(\frac{\partial\psi}{\partial x}\right)^2 + \left(\frac{\partial\psi}{\partial y}\right)^2$$
where the derivatives are computed spectrally: $\widehat{\psi_x} = ik_x\hat{\psi}, \quad \widehat{\psi_y} = ik_y\hat{\psi}$.
- **A3 — Vorticity-aligned.** Proportional to the SQG surface relative vorticity $\zeta = -\nabla^2\psi$:
$$\eta_{per} = \zeta$$
This perturbation is strong inside vortex cores.

1.1.2 Group B: Inertia–gravity waves

These perturbations mimic the balanced–unbalanced superposition common in altimetric observations.

- **B1 — Monochromatic plane wave.** A single plane wave with wavenumber $k_0 = 10$ and propagation direction $\theta = \pi/4$:
$$\eta_{per} = \cos\big(k_0(\cos\theta\, x + \sin\theta\, y)\big)$$
- **B2 — IGW wave packet.** A superposition of $N = 8$ plane waves with wavenumbers spanning mesoscale to submesoscale ($k_n \in \{5, 7, 10, 12, 15, 20, 25, 30\}$), uniformly spaced directions $\theta_n = n\pi/N$, and random phases $\varphi_n \sim \mathcal{U}(0, 2\pi)$:
$$\eta_{per} = \frac{1}{N}\sum_{n=1}^{N} \cos\big(k_n(\cos\theta_n\, x + \sin\theta_n\, y) + \varphi_n\big)$$

1.1.3 Group C: Stochastic noise

These perturbations are built from white noise $\hat{w}(\mathbf{k}) = \mathcal{F}\{w(x,y)\}$ where $w \sim \mathcal{N}(0,1)$, shaped spectrally to select different wavenumber bands.

- **C1 — Small Scale High k noise.** White noise restricted to $K > K_{cut}$
$$\eta_{\hat{per}}(\mathbf{k}) = \begin{cases} \hat{w}(\mathbf{k}), & K > K_{cut} \\ 0, & \text{otherwise} \end{cases}$$
Note that high-wavenumber correspond to small scale.
- **C2 — Large Scale Low k noise.** White noise restricted to $K < K_{cut}$, with the DC component zeroed:
$$\eta_{\hat{per}}(\mathbf{k}) = \begin{cases} \hat{w}(\mathbf{k}), & 0 < K < K_{cut} \\ 0, & \text{otherwise} \end{cases}$$

- **C3 — Red noise (K^{-2} energy spectrum).** White noise multiplied by K^{-1} in spectral space, producing an energy spectrum $\propto K^{-2}$:

$$\hat{\eta_{per}}(\mathbf{k}) = \frac{\hat{w}(\mathbf{k})}{K}, \qquad K > 0; \qquad \hat{\eta_{per}}(\mathbf{0}) = 0$$

1.1.4 Group E: SWOT-like instrument and geometry errors

These perturbations emulate error patterns characteristic of the SWOT satellite altimeter.

- **E1 — KaRIn-like anisotropic noise.** White noise shaped by an anisotropic power spectral density that decorrelates faster in the cross-track direction (y) than along-track (x). With decorrelation wavenumber $k_c = 10$:

$$\hat{\eta_{per}}(k_x, k_y) = \frac{\hat{w}(k_x, k_y)}{\sqrt{1 + (k_y/k_c)^2}}, \qquad \hat{\eta_{per}}(\mathbf{0}) = 0$$

  The amplitude transfer function suppresses high cross-track wavenumbers while leaving along-track wavenumbers unfiltered, reproducing the spectral anisotropy of KaRIn interferometer baseline errors.

1.2 Setup of the Benchmark

Fix the number of iteration after stop, compare the final loss and the inversion error.